

Step-by-step ZeroTier setup

1. Connect to the internet. Ensure that the Raspberry Pi Zero and any devices intended to access it are connected to the internet. A mobile Wi-Fi hotspot can be used for this setup.

2. Join a ZeroTier network.

- **Create a ZeroTier account.** Visit ZeroTier Central, create an account using an email address, and log in.

- **Set up a network.** Click on 'Create a Network' to generate a unique Network ID (a long hexadecimal string) that identifies the

virtual network.

3. Add and authorise devices on ZeroTier.

- **Connect devices.** Install ZeroTier One on each device and connect them to the network using the Network ID.

- **Mobile devices.** Install the ZeroTier One app, then enter the Network ID in the 'Join Network' section to establish a connection.

- **Raspberry Pi Zero.** Install and join the ZeroTier network using the following commands:

```
$> sudo apt install -y zerotier-one
[It may ask your sudo password]
```

To connect to the ZeroTier network here is the command:

```
$> sudo zerotier-cli join
xxxxxxxxxxxxxxxx [xxxx stands for the
network identifier]
$> sudo zerotier-cli info [provides info
about the network]
$> sudo zerotier-cli listnetworks
[provides the external IP address
of the Pi Zero]
```

4. Authorise devices. In ZeroTier Central, navigate to the network settings and tick the authorisation box next to each device's name. Once authorised, each device is assigned a unique ZeroTier IP address, which can be viewed in the ZeroTier Central dashboard.

5. Verify the connection.

- **Ping devices.** From the Pi Zero terminal, use the ping command to test connectivity with other devices on the ZeroTier network:

```
$> ping <ZeroTier-IP-of-other-device>
```

A successful response confirms that a secure connection has been established. This 'invisible pipeline' functions as a secure VPN tunnel, keeping communication private and accessible only to authorised devices.

6. Access the stream from anywhere.

- Open a browser on any authorised device and go to:

```
http://<ZeroTier-IP-of-Pi-Zero>:8081
```

to access the video stream.

- The stream loads seamlessly, accessible from anywhere in the world, as long as both devices are connected to the internet.

- **Secure connectivity.** Although communication takes place over the internet, it remains private; only devices within the ZeroTier network can access the stream.

- **Persistent connection.** The setup remains active and accessible across different locations, as long as the devices are online and connected to ZeroTier.

This approach provides a secure and cost-effective solution for streaming, surveillance, or remote monitoring, utilising only a Pi Zero and mobile data, without requiring third-party services.

Fig. 3 shows the ZeroTier Central dashboard with two active virtual networks (berapi-2 and berapi). It displays network IDs, subnets, the number of connected nodes, and creation dates. A notification indicates that 0 of 10 included devices have been used, with an upgrade option available to expand network capacity.

Fig. 4 shows screenshot of the ZeroTier One app connected to the virtual network berapi-2 with Network ID 856127940CBA92D1. The device is online and running version 1.14.0-2, confirming successful connection to the private network.

To ensure the Pi Zero automatically checks and connects to the ZeroTier network every time it boots, create a small shell script and set it up to run as a system service.

```
$> nano zerotier.sh
```

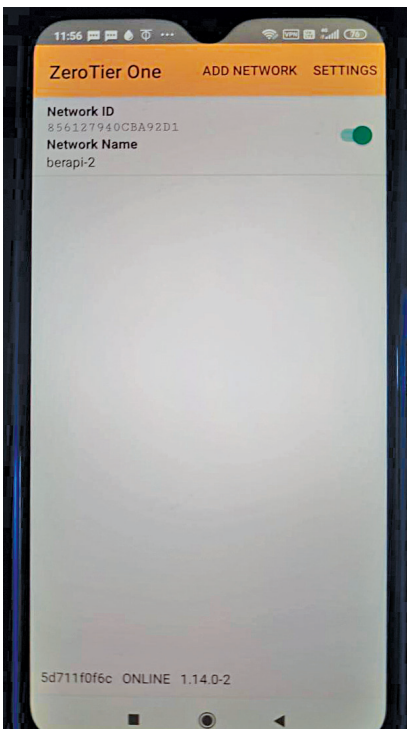


Fig. 4: Screenshot of ZeroTier

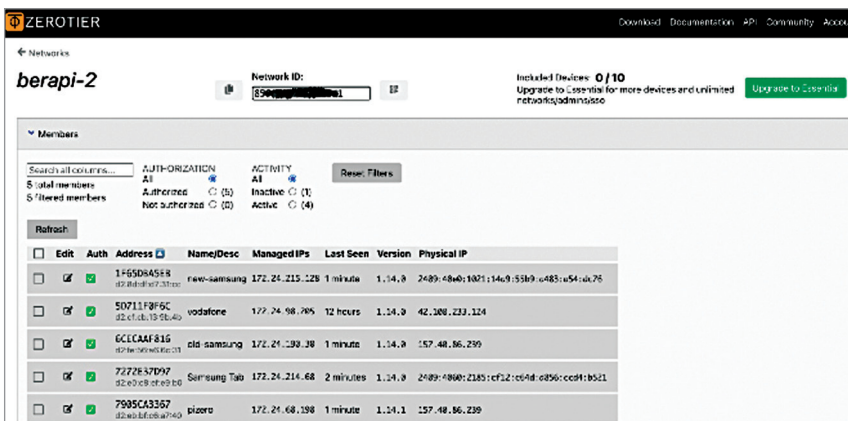


Fig. 5: Zerotier showing divides in the network